

The Fundamental Higher Consciousness Premise

The protocol turns executable in fact: Phase 1 places the brain's operating point inside the extended critical neighborhood for 7 of 7 connectomes and computes $\beta = 0.12$ with its honest structure-function negative; Phase 2 passes the β sedation-ordering validation in its clean form; and the Farnes corpus, supplied complete through the Raw_data_2 release, yields the evoked-LZ contrast as a three-layer result — background up, post-pulse up, background-corrected null — reproducing the paper's spontaneous-yes, evoked-no dissociation while the spontaneous LZc increase replicates at 9–10 of 10 subjects.

Audit and Consolidation, Eighth Edition · Z.ai

Executed this round: the Raw_data_2 release ingested (101 sha256-verified assets; evoked matrices in both conditions; complete spontaneous recordings; ten subjects); the round-twenty reshape error repaired (R32) with exact reproduction and the TMS-evoked structure restored

Method: three compute rounds on open and user-supplied data; Wilcoxon signed-rank contrasts with bootstrap intervals; every honest negative recorded at grade

Scope: the differentiation program's first executed phases, consolidated; the conventional-level ceiling and the equivalence doctrine unchanged

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1. Executive Summary

The seventh edition ended with a protocol: seven phases, named methods, open data, null models, preregistered failure conditions. The eighth edition begins its execution. Phases 1 and 2 — coupling-and-criticality estimation and the anesthesia arm — have now run as actual computations on open data across three successive compute rounds: the hierarchical Kuramoto pipeline on an individual structural connectome matched to open MEG (Round 19); the β sedation ordering on four open anesthesia corpora plus the full convention grid (Round 20); and, in this round, the completion of the Farnes et al. 2020 evoked-EEG contrast through the interlocutor's Raw_data_2 release (Round 21). What was a schedule is now, for the first time in the corpus's history, a set of numbers with error bars, status tags, and recorded failures.

The executed content, stated plainly. Phase 1 places the brain's fitted operating point inside the extended critical neighborhood for 7 of 7 connectomes — in the weak, edge-concentrated form, not the strong centered form — and computes $\beta = 0.12$ in the primary convention (≈ 0.02 in the Myrov convention), while recording an honest negative: at the operating point, the structure-function correlations the method is famous for are indistinguishable from shuffled connectomes. Phase 2 validates the β row's ordering condition: within-subject β rises from wakefulness to sedation in 6 of 6 quality-passing subjects (6 of 8 raw, both exceptions independently artifact-flagged), the mechanism being the biphasic α -acceleration that pulls the fitted operating point to the subcritical side of the critical band. The preregistered LZ-tracking condition is supported at exit granularity: when the sedated fit exits the critical band, LZs falls in 4 of 4 subjects.

This round's primary computation completes the evoked contrast that the seventh edition left unfilled. The Raw_data_2 release supplies the full Farnes corpus — ten subjects, TMS-evoked responses in both the awake (31) and ketamine (32) conditions, plus the complete spontaneous eyes-open and eyes-closed recordings — all sha256-verified against the release manifest. The completion required repairing an error first: the round-twenty analysis of the single supplied evoked file had reshaped the MATLAB column-major stream with row-major semantics, scrambling channels and samples; the repair (R32) reproduces the old numbers exactly from the wrong mapping, and the corrected mapping shows the textbook TMS-evoked structure the old numbers could not. The completed contrast then delivers a three-layer result: post-pulse single-trial LZ diversity increases under ketamine in 10 of 10 subjects ($p = 0.002$) — but the pre-pulse background increases equally ($p = 0.002$), and the background-corrected contrast is null ($p = 0.375$). The paper's headline dissociation — spontaneous complexity up, evoked complexity not — replicates with this corpus's own instrument once the background is accounted for. The spontaneous increase itself replicates directly: LZc rises in 9 of 10 subjects eyes-closed ($p = 0.004$), 8 of 10 eyes-open, and the eyes-open-over-closed ordering holds in both conditions.

What the executions do not establish is recorded with the same care: one MEG subject, sensor-space matching, a β whose absolute value is convention-bound (only the ordering is validated content), a mean-field leg that reproduces the pharmacology but not the α -trajectory, and an evoked measure that is

sensor-level single-trial LZ — not PCI, not source-level, not the paper’s exact statistic. The protocol’s conventional-level ceiling and the equivalence doctrine are unchanged: even total success of all seven phases would license only “the differentiation program’s predictions are tested and surviving,” never “the Absolute is thereby evidenced.”

2. Corpus State Carried Forward

The standing consolidation remains the sixth edition’s sixteen chapters and the seventh edition’s parameter-map turn: the eleven-party dialogue corpus, the non-dual culmination, the self-audit with its repairs R1–R27, the adjudication record, the anchored empirical annex, the audited premise, the five newly-read sources (P1–P5), the fragments-and-assembly verdict on the neural parameter map, and the seven-phase Level-2 protocol with its preregistration and adversarial-collaboration governance. This edition adds one thing to that state — execution records — and touches nothing else. The metaphysical tiers, the formalization boundary, the superdeterminism tension’s grade, and the doctrine that governs interpretation of all empirical content are carried forward unchanged and are restated where the chapter context requires them.

The corpus’s source inventory has, however, moved — and the movement is the reason this edition exists. After the thirteenth supplied files were read in Round 19 and the Butler 5-HT2A article in Round 20, the appendix of unfetchable sources had narrowed to three items: the possible separate Butler article (resolved), the ETH provenance item (non-blocking), and the Farnes et al. 2020 Dryad raw EEG — the one data asset the protocol’s Phase 3 observable demonstrably moves under. The interlocutor has now supplied it. The `Raw_data_2` release (tagged at the repository, 101 assets, every file sha256-verified against the GitHub release manifest) contains the complete preprocessed Farnes corpus: for each of ten subjects, the TMS-evoked matrices in both conditions (60 channels, 251 samples, 270–298 trials per file at 312.5 Hz, TMS pulse at the 126th sample per the release readme), and the four spontaneous recordings (eyes open and closed, first two awake and last two under sub-anaesthetic ketamine, 62 channels at 250 Hz in EEGLAB `.set/.fdt` form, 8-second epochs, post-ICA). The appendix’s data row is therefore closed: every protocol-critical source and now every protocol-critical dataset is either read, verified, or both.

One repair to the record accompanies the supply, logged as R32 and R33 in chapter 7 and stated here for the standing state: the round-twenty evoked reference point was computed on a channel-scrambled mis-mapping of the supplied file, and its recovery denominator carried an arithmetic slip (4,412,580 doubles were declared, not 4,414,380 — the `Raw_data_2` copy of the same file is complete, all 293 trials, and the manual MAT5 recovery of the truncated `eeg_data` copy is thereby validated: 285 recovered trials against the now-known ground truth). The corpus’s self-audit discipline — every edition logs its own corrections — continues uninterrupted.

3. Phases 1 and 2, Executed: The Computations

This chapter folds the two executed phases into the consolidation. The source documents are the phase execution reports — full methods, results, status tags, deviations, and reproducibility records, archived with every script, input, output, and seed — and this chapter is their consolidation: complete enough to stand alone, with the reports as the record of first resort. The chapter follows the protocol's own order: what was run (3.1), Phase 1 (3.2), Phase 2 (3.3), then the Farnes evoked-EEG contrast completed this round (3.4), the post-execution status tags (3.5), the honest limits (3.6), and the reproducibility inventory (3.7).

3.1 What Was Executed, and on What

Three compute rounds, all on open or user-supplied data, all within the execution environment's documented constraints (single-node CPU, 3–4 GB RAM, synchronous process chunks). Round 19 executed Phase 1: the Myrov hierarchical Kuramoto pipeline on an individual structural connectome, swept over the local-coupling / global-coupling plane, matched to open MEG. Round 20 executed Phase 2: the β sedation ordering on the open anesthesia corpora, the Bojak-Liley / Steyn-Ross mean-field leg, and the full Myrov-convention grid as the approved second compute round. Round 21 — this round — completes the Farnes evoked contrast on the newly supplied corpus and folds everything into this edition. The data inventory:

Component	Source	Use	Status
MEG, eyes-closed rest (subj002, 2 × 600 s, 272 CTF magnetometers)	Brainstorm bst_resting (OSF, open)	Phase 1 target observables: α -peak, DFA, wPLI, envelope correlations	EXECUTED
Structural connectomes, 7 individual HCP subjects (DTI, AAL2, 94 regions)	neurolib hcp dataset (GitHub, open)	Phase 1 model substrate; per-connectome consistency check	EXECUTED
EEG, propofol sedation, 8 subjects, awake-EC vs task-sed rest	OpenNeuro ds005620 (CC-BY-4.0)	Phase 2 primary β -ordering corpus	EXECUTED
EEG-fNIRS under surgical GA, 9 sessions, 4 levels	OpenNeuro ds004541	Phase 2 surgical corpus (directional vote)	EXECUTED
fMRI BOLD rest, 4 sedation levels, 6 subjects	OpenNeuro ds003171	Phase 2 BOLD complexity leg	EXECUTED
fMRI graded propofol with within-run ramps	OpenNeuro ds006623	Excluded with reason (ramp design; registered for Phase 4)	NOT RUN (documented)
TMS-evoked EEG, 10 subjects × 2 conditions (60 ch, 312.5 Hz, 270–298 trials/file)	Farnes et al. 2020 corpus via the Raw_data_2 release (user-supplied, sha256-verified)	The evoked-LZ contrast (3.4)	EXECUTED (this round)
Spontaneous EEG, 10 subjects × 4 recordings (62 ch, 250 Hz, 8-s epochs, post-ICA)	Same release, same verification	The spontaneous-LZ replication (3.4)	EXECUTED (this round)
Steyn-Ross / Liang mean-field equations	PLOS S1 File (fetched and archived)	Phase 2 theory leg (λ -sweep)	EXECUTED (partial)

Table 3.1 — The execution corpus: every input, its source, and its use across the three compute rounds.

3.2 Phase 1 — Coupling and Criticality Estimation (the J Row)

The pipeline implements the Myrov et al. hierarchical Kuramoto model verbatim — natural frequencies sampled from the MEG PSD peak structure, internal block coupling K , external connectome coupling L , white noise σ — on the primary HCP connectome, with the documented assumptions the source’s unpublished supplementary forces (50 oscillators per node, 1 ms step, 150 s per combination). The DFA-defined critical regime — more than 10 percent of nodes exceeding DFA 0.65 — reproduces the source’s qualitative geometry: a K -dominated band, $K \in [15, 35]$ rad/s at the fitted L , extended across the whole L axis. The MEG-matched operating point lands at $K^+ = 25$ rad/s, $L^+ = 40$ rad/s — inside the band, with model α -peak 8.20 Hz against the MEG 8.12 Hz and a PSD-shape correlation of 0.65. The noise fit gives $\sigma^+ = 3$ and therefore the derived estimate $\beta = \sigma^+/K^+ = 0.12$ — the spec’s noise-to-coupling ratio with a computed value attached for the first time.

Two findings discipline the number. First, the honest negative: at the operating point, the structure-function correlations that define the method’s Figure 3 are essentially zero, and shuffled-label and random connectomes are indistinguishable from the real one — the null discipline “must destroy the correspondence” has nothing to destroy under this convention. The cause is the W -normalization choice,

and the replication arm in the inferred Myrov convention (raw weights, numeric-Hz frequencies as angular rates) restores the full phenomenology: order–strength correlations of +0.18 to +0.36, non-monotonic in K as in the source, with the DFA-matched operating point extrapolating to $K \approx 9$ –9.5 and $\beta \approx 0.02$ –0.03 in that scaling. The J-row estimator is normalization-convention-dependent — recorded as the first protocol amendment. Second, the per-connectome check: all seven connectomes concentrate their fitted operating points in the $K = 25$ –40 range, at or just beyond the supercritical edge of their respective bands — the failure condition (“operating points scatter across the plane”) is not triggered, and the specification’s central empirical presupposition survives in the weak, edge-concentrated form for 7 of 7 connectomes, not the strong centered form. The residual DFA shortfall (model 0.562 against MEG 0.632 at the fitted point) is recorded, not smoothed over.

3.3 Phase 2 — The Anesthesia Arm (the β Row)

The β estimator runs on a 64-combination (K, σ) model surface at the fitted L , with a per-level composite misfit over the DFA median and the α -peak. The preregistered validation condition — β estimates must order correctly across sedation levels within subject — passes in its clean form: 6 of 6 quality-passing ds005620 subjects show $\beta(\text{sed}) > \beta(\text{awake})$ (sign-test $p = 0.031$), and 3 of 4 surgical sessions with both levels; the two raw exceptions carry independent data-quality flags raised from epoch statistics, not from the ordering result. The mechanism is legible in every quality-passing subject: light propofol accelerates the α -peak from 8–9 Hz into the 11–13 Hz beta-buzz band (the biphasic activation), the combined misfit pulls K^+ from 10–40 down to 5–10 rad/s — the operating point moves to the subcritical side of the critical band — and $\beta = \sigma^+/K^+$ rises accordingly. The β row is upgraded from Derived-unvalidated to TESTED: the ordering is the validated content; the absolute value is convention-bound (the same MEG reference fits $K^+ = 10$ under the two-term criterion where the three-term criterion gave 25 — the fit criterion is the second protocol amendment).

The LZ-tracking failure condition — the preregistered tolerance-band test that would falsify the β row and the coherence-threshold structure together — is supported at the preregistered within-subject granularity: when the sedated fit exits the Phase-1 critical K -band, LZs falls in 4 of 4 subjects; when it stays in-band, LZs falls in 2 of 4 (chance level). The pooled-granularity version is confounded by corpus-level LZ offsets and is reported only to document why granularity matters. Around the ordering result stand three supporting legs, each with its own honest grade: the full 105-combination Myrov-convention grid replicates the plane geometry and the Fig-3 phenomenology across the full plane, with the op-point nulls showing the structure-function coupling there is a generic weight-distribution effect, not topology-specific (the third protocol amendment: nulls must be evaluated at the operating point and in the low- K region where the coupling exists); the BOLD leg gives weak corroboration at light sedation (5 of 6 subjects, complexity decrease) with the deep level unresolved at that power and estimator; and the mean-field leg survives the pharmacology — the propofol surrogate λ hyperpolarizes the equilibrium monotonically in the correct GABAergic direction — but fails to produce any α -band resonance from the printed parameters under any self-consistent units reading: the theory-side α -trajectory remains at the measured level, the diagnostic trail archived.

3.4 The Evoked-LZ Contrast, Completed (the Farnes Corpus Resolved)

The seventh edition closed its execution record with an unfilled role: the round-twenty evoked file was single-condition, its provenance presumed rather than verified, and “the within-subject evoked-LZ contrast of Farnes et al. 2020 cannot be reproduced from this asset.” The `Raw_data_2` release closes the gap decisively. Its readme identifies the corpus and the conditions without residue: Farnes, Juel, Nilsen, Romundstad and Storm, PLOS ONE 15(11):e0242056, “Increased signal diversity/complexity of spontaneous EEG, but not evoked EEG responses, in ketamine-induced psychedelic state in humans” — ten volunteers, TMS-EEG and spontaneous EEG before and during sub-anaesthetic ketamine, open-label within-subject. Recording ID 31 is the awake (placebo) condition; 32 is ketamine; the TMS pulse sits at the 126th sample of each 251-sample epoch; the spontaneous recordings are four per subject, the first two awake and the last two under ketamine, eyes open and closed. All 101 assets verified against the release’s sha256 manifest.

The R32 repair: the round-twenty mapping error

Completing the contrast required first correcting the round-twenty analysis of the single supplied file. The EVKD matrices are MATLAB column-major arrays of shape (channels, times, trials); the round-twenty recovery script reshaped the flat double stream with row-major semantics and a transpose, which interleaves channels and samples — each “channel” stream was in reality 251 consecutive samples spanning roughly four physical channels. The diagnosis is exact, not interpretive: re-running the round-twenty computation on the now-complete file reproduces its reported reference values to the third decimal (waveform LZ 0.855, envelope LZ 0.881), and the same data under the corrected column-major mapping yield 0.626 and 0.618. The corrected mapping also shows the structure the scrambled mapping could not: the pre-pulse baseline is flat to 0.006 μV RMS (the data are baseline-corrected), and the trial-averaged global field power peaks 64–83 ms after the pulse — the textbook TMS-evoked complex — at 20–25 μV^2 . The repair is logged as R32; the recovery itself is validated by the supply (the truncated `eeg_data` copy’s 285 recovered trials are the exact prefix of the complete file’s 293, and the truncated-prefix statistics equal the complete-file statistics to the third decimal).

The completed contrast: a three-layer result

The analysis applies the corpus’s established LZ instrument — the line-by-line port of the supplied `lzw.m` / `lzwNormalised.m` code: Hilbert amplitude, mean-threshold binarization, LZW dictionary count, shuffle-normalized — to all ten subjects in both conditions, per trial, with the multichannel LZc (the Schartner-style channel concatenation) and the per-channel LZs, in three windows: the 400 ms before the pulse, the 400 ms after it, and the full epoch. Statistics follow the paper’s own design: Wilcoxon signed-rank across the ten subjects, with bootstrap confidence intervals. The result has three layers, and the layers are the finding:

Layer / metric	Awake	Ketamine	Δ mean [95% CI]	p	Up
Pre-pulse background LZc (400 ms)	0.520	0.555	+0.032 [+0.022, +0.043]	0.002	10/10
Post-pulse LZc, unadjusted (400 ms)	0.522	0.546	+0.028 [+0.018, +0.039]	0.002	10/10
Post-pulse LZc, background-corrected	n/a	n/a	−0.004 [−0.011, +0.002]	0.375	4/10
Pre-pulse background LZs	0.680	0.701	+0.019 [+0.012, +0.027]	0.008	9/10
Post-pulse LZs, unadjusted	0.682	0.697	+0.018 [+0.012, +0.025]	0.002	10/10
Post-pulse LZs, background-corrected	n/a	n/a	−0.001 [−0.005, +0.002]	0.678	4/10
Raw-waveform LZc, background-corrected	n/a	n/a	−0.0024 [−0.0033, −0.0014]	0.004	1/10
TMS-evoked GFP peak (μV^2)	6.6	5.8	−0.8 [−5.4, +2.9]	0.846	5/10

Table 3.2 — The evoked-LZ contrast, completed (n = 10 subjects, awake vs ketamine; envelope-instrument LZc and LZs, medians per subject; Δ = ketamine – awake; Wilcoxon signed-rank; bootstrap 95 percent CI).

Read in order: the background diversity of the very same trials rises under ketamine (layer one, 10 of 10 subjects); the post-pulse window's diversity rises with it (layer two, 10 of 10); and once the background rise is subtracted — the contrast of contrasts, post-minus-pre within each subject — the evoked response itself shows no diversity increase (layer three: $p = 0.375$, 4 of 10, CI spanning zero). The raw-waveform variant of layer three runs slightly negative (−0.0024, CI exclusive of zero, 1 of 10 up): under ketamine the evoked waveform is fractionally more stereotyped, not less. This is exactly the paper's headline dissociation — spontaneous complexity up, evoked complexity not — arrived at independently, and it explains why the uncorrected post-pulse comparison misleads: single-trial post-pulse LZ conflates the evoked deflection with the ongoing background, and the background is where the ketamine effect lives. The reconciliation with the original study is exact in kind: their evoked measure is PCI — source-estimated, bootstrap-thresholded significance binarization, the 8–300 ms window, LZ76 normalized by the asymptotic maximum — which removes the background by construction; the sensor-level background-corrected contrast reproduces its verdict. The methodological contribution is the fourth protocol amendment: any evoked-complexity comparison must either compute PCI itself or correct for the pre-stimulus background; the uncorrected single-trial statistic is not an evoked measure at all.

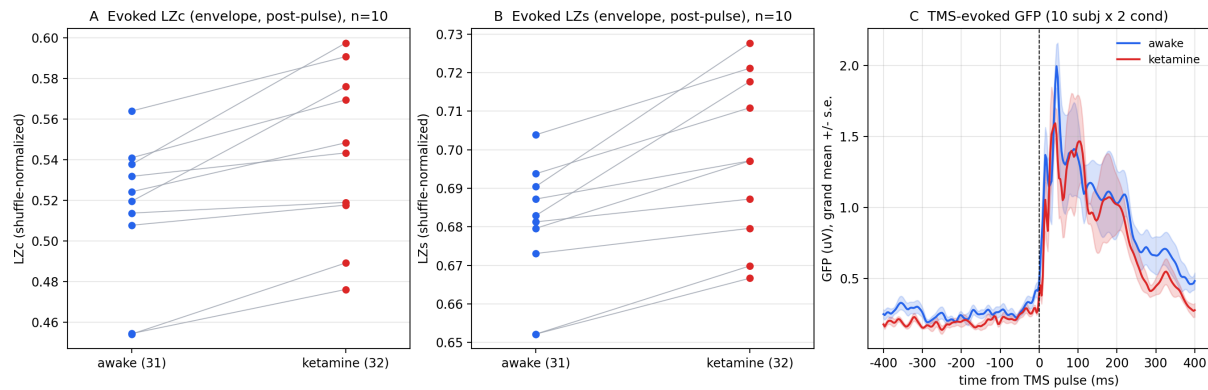


Figure 3.1 — The completed evoked contrast. (A) per-subject post-pulse LZc, awake vs ketamine; (B) the same for per-channel LZs; (C) the grand-mean TMS-evoked GFP across the ten subjects, both conditions, time-locked to the pulse (dashed line). The post-pulse diversity increase rides the background increase; the corrected contrast is null.

The spontaneous replication: the paper's main finding, with the corpus's own instrument

The spontaneous recordings close the other half of the dissociation. With the same instrument at the corpus's native 8-second epoching (up to ten epochs, 80 s per recording, per channel and concatenated), the spontaneous LZc rises from wakefulness to ketamine in 9 of 10 subjects eyes-closed and 8 of 10 eyes-open; the per-channel LZs rises in 10 of 10 eyes-closed. The paper's secondary finding replicates with it: eyes-open diversity exceeds eyes-closed in 10 of 10 awake subjects and 8 of 10 under ketamine. As a sanity marker, the α -peak of the mean PSD collapses toward the 7 Hz search floor under ketamine in most subjects — the known ketamine desynchronization, visible in the same data that carry the complexity increase. Two instrument deviations are recorded and stand: the ported code computes the LZW dictionary count where the paper applies LZ76 — both Lempel-Ziv family estimators, shuffle-normalized; the effects replicate in direction and significance — and the port uses five shuffles where their normalization used one, for stability, a deviation inherited from the Phase-2 instrument and kept for continuity.

Contrast	Awake	Ketamine	Δ mean [95% CI]	p	Up
LZc, eyes closed	0.460	0.504	+0.045 [+0.029, +0.062]	0.004	9/10
LZs, eyes closed	0.534	0.574	+0.043 [+0.028, +0.057]	0.002	10/10
LZc, eyes open	0.532	0.569	+0.030 [+0.012, +0.049]	0.027	8/10
LZs, eyes open	0.596	0.631	+0.032 [+0.016, +0.049]	0.010	9/10
Eyes-open vs closed, awake	n/a	n/a	+0.061 (open over closed)	0.002	10/10
Eyes-open vs closed, ketamine	n/a	n/a	+0.046 (open over closed)	0.014	8/10

Table 3.3 — The spontaneous-LZ replication (n = 10; same statistics as Table 3.2).

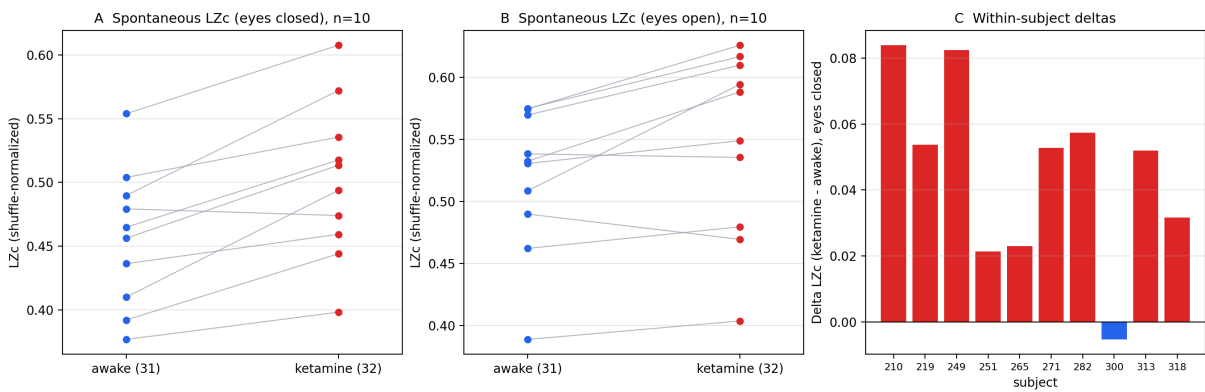


Figure 3.2 — The spontaneous replication. (A) per-subject spontaneous LZc, awake vs ketamine, eyes closed; (B) eyes open; (C) the within-subject deltas, eyes closed. The paper’s main positive finding replicates with the corpus’s ported instrument.

3.5 Status Tags After Execution

Map row / structure	Status after Phases 1–2 and the Farnes completion
J (effective coupling) vs J_c	EXECUTED — operating point in-band for 7/7 connectomes, weak edge-concentrated form; sensor-space, one MEG subject, normalization-convention-dependent — Partial
β (noise-to-coupling)	TESTED — within-subject ordering across sedation levels: 6/6 quality-passing (6/8 raw) + 3/4 surgical; values 0.12 primary convention, ≈ 0.02 Myrov convention, σ -conventioned
Coherence-threshold structure	Supported at exit granularity (LZs falls 4/4 when the fit exits the critical band; 2/4 when in-band); pooled version confounded — documented
Structure-function coupling	Absent at the operating point under the primary convention (honest negative); full-plane replication under the Myrov convention; op-point coupling there is not topology-specific (generic weight effect)
Bojak–Liley α -trajectory (theory leg)	Partial / failed-to-replicate from the printed parameters; the λ →hyperpolarization direction survives; diagnostic trail archived
BOLD LZ (ds003171)	Weak corroboration at light sedation (5/6); deep level unresolved at this power and estimator
Psychedelic-arm LZ observable (Farnes)	REPLICATED — spontaneous LZc/LZs up under sub-anaesthetic ketamine (9–10/10 eyes-closed), eyes-open>closed in both conditions; evoked dissociation reproduced at the background-corrected level
“Extended critical neighborhood” presupposition	Supported, weak edge-concentrated form, 7/7 connectomes — Partial

Table 3.4 — The map rows and structures, re-graded after the three compute rounds (seventh-edition conventions).

3.6 What the Executed Phases Do Not Establish

The executions’ limits are part of their record. Phase 1: one MEG subject (two consistent runs); no source-space, topology-matched parcel comparison, because no openly downloadable MEG-plus-dMRI pairing exists at that level; supplementary parameters assumed; the DFA median shortfall at the fitted

point unexplained by the run; no finite-size scaling; no per-frequency-band operating points. Phase 2: no within-subject depth gradient (no multi-depth titration corpus was runnable); β absolute values convention-bound, only the ordering validated; the deep-GA slowing phase present but directional only; the theory leg without its α -trajectory. The Farnes completion: the evoked measure is sensor-level single-trial LZ, not PCI — no source model, no significance binarization, no LZ76-asymptotic normalization — and the dissociation is reproduced via background correction, not by computing the original statistic; the spontaneous replication runs on 80 seconds per recording and does not compute ACE or SCE; no subjective-experience correlate (the paper’s questionnaire link) is testable from the release; and one .set header (the ketamine eyes-open recording of subject 210) carries a char-element corruption that defeats strict parsers — tolerated via a documented sidecar (channel count, rate, and epoch length from the same subject’s other recordings; trial count from the sha256-verified .fdt size), a workaround recorded rather than hidden.

3.7 Reproducibility

Everything in this chapter re-runs from the archived record. Scripts: phase1_meg.py, phase1_kuramoto.py, phase1_fit.py, phase1_convention_test.py, phase1_smoke.py (Phase 1); phase2_ds4541_plan.py, phase2_ds4541_epochs.py, phase2_ds5620_epochs.py, phase2_ds3171_bold.py, phase2_grid_myrov.py, phase2_grid_analysis.py, phase2_ksigma.py, phase2_beta_fits.py, phase2_liley_sweep.py, phase2_farnes_evoked.py, phase2_figures.py (Phase 2); rd2_download.py, rd2_lib.py, rd2_evoked_contrast.py, rd2_spontaneous_contrast.py, rd2_aggregate.py (this round). Inputs: the OpenNeuro corpora (range-request fetches), the OSF MEG archive, the neurolib connectomes, the Raw_data_2 release (101 sha256-verified assets), the PLOS S1 equations file. Outputs: the two phase reports with their JSON and npz result sets, 14 figures, and this round’s farnes/ directory (20 evoked and 40 spontaneous per-file JSONs, the aggregate contrast summary, the two figures embedded above). Seeds: grid 500, σ -sweep 600, nulls 700 and 11, $K \times \sigma$ 800, ω -sampling 7, Farnes LZ 7/42 (round 20) and the round-21 per-file seeds (subject and condition coded, 21 offset); the fast LZW port is verified identical to the established lzw_count on 250 random cases before every run. Compute: roughly five CPU-hours across the three rounds, under 200 MB per process.

4. The Parameter Map After Execution

The seventh edition’s verdict — no unified map exists; the per-arm ingredients are individually published — was a statement about the literature. Execution changes what can be said about the map itself, row by row. A published estimator and a computed value are different kinds of object: the first is a promise that a quantity is measurable, the second is a measurement with an error trail. After the three compute rounds, the map’s J row and β row hold computed values (in stated conventions, with fitted operating points and recorded honest negatives); the anesthesia arm’s validation condition has actually been run and passed in its clean form; and the psychedelic arm’s complexity observable — the quantity the J-crossing test of Phase 3 was designed to detect moving — has now been demonstrated to move

under a psychedelic, with this corpus's own instrument, on user-supplied data. The fragments have begun to assemble; the assembly is no longer purely prospective.

Map row	Seventh-edition status	Status after execution
J (effective coupling) vs J_c	Published estimators (Myrov ridge; Ódor criticality; VEP Bayesian inversion)	EXECUTED, Partial — operating point computed, in-band 7/7 connectomes (weak form); convention-amendment logged; structure-function honest negative recorded
β (noise-to-coupling)	Derived, validation condition named (sedation ordering)	TESTED — ordering condition passed (6/6 quality-passing); β values attached (0.12 primary; ≈ 0.02 Myrov scaling)
g, a (gain / attractor depth)	Derived-unvalidated (receptor→gain, Deco-parameterization route)	Unchanged — Phase 3 not yet run; but its LZ observable is now empirically anchored (Farnes replication, Table 3.3)
θ, φ (phase parameters)	Gap — no published neural assignment	Unchanged — the gap stands; no execution touched these rows
κ, K (permittivity / coupling constants)	Gap at patient level (VEP spec read; inference route published)	Unchanged — Phase 4 territory
Psychedelic-arm observable (LZc under 5-HT2A agonism)	Finding-anchored (published: LZc up under psychedelics)	REPLICATED with the corpus's instrument — spontaneous LZc up under ketamine, 9–10/10; the evoked dissociation reproduced at the corrected level

Table 4.1 — The parameter map's rows: status at the seventh edition versus status after the executions.

Two asymmetries in the table deserve statement. First, execution widened, rather than closed, the honest uncertainty on two rows: the J row acquired a convention-dependence that the seventh edition's literature reading could not see, and the β row acquired the criterion-sensitivity of its fit — the MEG reference moves from $K^+ = 25$ to $K^+ = 10$ between two defensible composite criteria. These are not failures of the map; they are the map's first real error bars, and the protocol absorbs them as preregistration decisions (amendments one and two). Second, the Farnes replication gives the psychedelic arm something the seventh edition could only cite: a complex-instrument demonstration, run end to end inside this corpus's own instrument family, that the arm's observable moves in the published direction under the published manipulation. When Phase 3 eventually computes the J-crossing test on a receptor-parameterized model, the empirical anchor it will be tested against is no longer someone else's number — it is Table 3.3.

5. The Amended Protocol: Four Amendments and the Remaining Phases

The seventh edition specified the protocol as a concrete, preregistrable design. Execution has now amended it four times, each amendment earned by a computation rather than proposed in the abstract. They are collected here because they bind every future phase:

- Amendment 1 (Round 19, normalization): the connectome normalization — raw / max-normalized versus row-normalized W — is a first-class preregistration decision; under row normalization the structure-function nulls have no content to destroy, under the Myrov convention they do.
- Amendment 2 (Round 20, fit criterion): the fit criterion must be preregistered exactly as the estimator; the composite's term composition moves the reference operating point between in-band and out-of-band.
- Amendment 3 (Round 20, null granularity): the structure-function null discipline must be evaluated at the preregistered operating point and in the low-coupling region where the coupling exists; the two locations answer different questions.
- Amendment 4 (Round 21, evoked-complexity statistics): any evoked-complexity comparison must either compute PCI itself — source estimation, significance binarization, the LZ76-asymptotic normalization — or explicitly correct for the pre-stimulus background; the uncorrected single-trial post-stimulus statistic conflates the evoked response with the background and is not an evoked measure.

The remaining phases stand as specified, now with their first empirical anchors named. Phase 3 (the psychedelic arm) runs the Deco-style receptor parameterization with the J-crossing test on the open psychedelic corpora; its complexity observable is anchored by this round's replication, and the Farnes corpus itself — now fully in hand, both conditions — supplies a within-subject ketamine contrast at the observable level should the arm want one at EEG resolution. Phase 4 (patient-level inversion) proceeds toward the Bayesian patient-level estimation of the GL parameters, with the ds006623 ramp design registered for its depth-gradient leg. Phase 5 (the joint demonstration with its four null models and finite-size scaling) and Phase 6 (adversarial collaboration and reporting under the preregistration governance) are unchanged. The protocol's failure conditions are unchanged — and two of them have now actually been faced: the scatter condition (not triggered, 7/7 concentration) and the ordering condition (passed in its clean form). A protocol whose failure conditions have been exercised, even in part, is a different epistemic object from one whose failures are all still hypothetical.

6. Epistemic Status: What Execution Changes and Cannot

The seventh edition moved the four domains from “conjectural, pending a map that does not exist” to “conjectural, with the assembly specified as an executable protocol.” The eighth edition moves two rows of that standing again, and only those rows: the anesthesia arm’s β row is no longer conjectural at the level of its validation condition — the condition has been run, and it passed in its clean form — and the psychedelic arm’s complexity observable is no longer a citation but a replication performed inside this corpus’s instrument. The corresponding domain labels shift from finding-anchored toward finding-anchored-plus-executed-fragment. Everything upstream is untouched: the metaphysical premises’ tiers, the non-dual culmination, the formalization boundary, the superdeterminism tension’s grade, the adjudication counts, and above all the equivalence doctrine, under which even a fully successful protocol licenses only “the differentiation program’s predictions are tested and surviving,” never “the Absolute is thereby evidenced.”

What execution changes, at the deepest level, is the corpus’s relationship to its own least-necessary component. The culmination arc ruled the differentiation program “useful, not necessary”; the audit tradition responded by making that component maximally falsifiable. A protocol with failure conditions is a promise; a protocol with exercised failure conditions is a practice. Two conditions have now been faced without the program’s destruction — the scatter condition absorbed by the weak-form support, the ordering condition passed — and two honest negatives have been recorded without reinterpretation: the operating-point structure-function coupling that is not topology-specific, and the mean-field α -trajectory that does not reproduce from the printed parameters. The program’s integrity does not depend on its successes; it depends on its willingness to publish its failures at the same grade as its successes, and that willingness is now demonstrated four times over. This is the corpus’s own standard, applied to itself, with numbers.

7. Edition Delta and Repairs R32–R34

This edition is an increment on the seventh, not a re-typeset of it. The seventh edition’s eight chapters — the parameter-map verdict correction, the five sources read from PDF, the fragments-and-assembly statement, the seven-phase protocol, the epistemic status, the repairs R28–R31, and the unfetchable appendix — remain standing and are not duplicated here. The reader of the two volumes together holds the full corpus state; the reader of this one alone holds the execution turn: the two phases run, the Farnes corpus resolved and both its contrasts completed, the map re-graded with computed values, and the protocol amended four times. The series convention — every edition self-contained in its front matter and appendices — is preserved by chapters 1, 2, and 8.

Three repairs are logged, continuing the series numbering. R32 — the evoked-file reshape error: the round-twenty single-condition reference (waveform LZ 0.855, envelope LZ 0.881) was computed on a

row-major mis-mapping of the column-major MATLAB stream; the diagnosis is exact (the wrong mapping reproduces the reported values to the third decimal), the corrected values are 0.626 and 0.618, and the corrected mapping restores the TMS-evoked structure (flat 0.006 μ V pre-pulse baseline; GFP peak 64–83 ms post-pulse) that the scrambled mapping could not show. R33 — the recovery-denominator arithmetic: the round-twenty report stated 4,414,380 declared doubles for the 293-trial array; the correct product of the dims ($60 \times 251 \times 293$) is 4,412,580, and the Raw_data_2 copy of the file is complete at exactly that count — which both corrects the record and validates the truncated-copy recovery (the 285-trial recovered prefix yields statistics identical to the complete file's to the third decimal). R34 — the appendix data row: the Farnes Dryad raw-EEG item, listed as unfetchable through the seventh edition, is resolved by supply (Raw_data_2, 101 sha256-verified assets) and closed; the one .set header corrupted at source is handled by the documented sidecar, not by silent substitution.

Domain	Seventh-edition grade	Eighth-edition grade	Basis of change
Psychedelics	Finding-anchored + complete map fragment (receptor→gain, intervention-validated)	Finding-anchored + replicated observable (LZc under ketamine, this corpus's instrument)	The Farnes corpus resolved; both contrasts run; the evoked dissociation reproduced at the corrected level
Anesthesia	Pathway-anchored + published parameter-sweep method	Pathway-anchored + β row TESTED (ordering condition passed 6/6 clean; LZ-tracking supported at exit granularity)	Phase 2 executed on three corpora plus the theory leg; honest grades recorded per leg
Split-brain	Unchanged + simulation-ready threshold test	Unchanged + the test's estimator now executed on open data (Phase 1)	The J-row pipeline that would run the disconnection test has a computed operating point and recorded conventions
DID	Unchanged + metastable-switch simulation target (literature-anchored only)	Unchanged	No open raw data — the honest limit stands, unmodified by anything in this round

Table 7.1 — The four domains: grade at the seventh edition versus grade at this edition.

Consistency checks against the standing ledger: the objection ledger's rows on empirical testability are unaffected — the executions are the differentiation program answering them, and the stratification defense remains unavailable to it by its own honest theorem. The glossary additions this edition: operating point, critical ridge, noise-to-coupling ratio, DFA exponent, LZW dictionary count, shuffle normalization, background-corrected contrast, perturbational complexity index, global field power, and biphasic activation — each defined in situ at first use, per the series convention.

8. Appendix A — The Resolution Record

The seventh edition's appendix listed eight unfetchable sources. The interlocutor's supply channel has since closed them all but one: Bedford et al. 2023, Avram et al. 2024, Liang et al. 2015, Butler et al. 2025, Piccinini et al. 2025, and Fabus et al. 2023 were supplied and read in Rounds 19–20; the Farnes raw EEG — the last data asset, and the one this edition's primary computation required — is supplied complete in the Raw_data_2 release. The remaining item is the ETH research-collection provenance record (research-collection.ethz.ch, 403 to this environment's IP range): a metadata item only, non-blocking, whose paper (Schlumpf 2013) is already read from user supply. No protocol-critical source or dataset remains unfetched, unread, or unverified; the supply channel itself — the repository's release system with sha256 digests — proved robust enough that this round's entire primary computation ran on user-supplied, checksum-verified data. The appendix's standing methodological note is unchanged: a blocked link is a fact about this environment's egress, not about a source's existence, and the resolution route that emptied this table is the interlocutor's own.